

Dynamic Pricing System with Neural Networks, LLM & RAG - Case Study

1. Executive Summary

The Dynamic Pricing System is an AI-powered solution that predicts demand, recommends optimal prices, analyzes competitors, and maximizes revenue. The system combines Machine Learning, Neural Networks, LLM capabilities, and a RAG component with an interactive Streamlit/Gradio interface.

2. Problem Statement

Traditional static pricing fails to adapt to market demand, weather, seasonal changes, and competitor pricing. Businesses require dynamic pricing strategies to maximize revenue and customer satisfaction.

3. Objectives

- Predict future demand accurately.
- Recommend optimal prices in real time.
- Analyze competitor pricing trends.
- Optimize revenue and profitability.
- Integrate Neural Networks for advanced predictions.
- Integrate LLM + RAG for intelligent user interaction.
- Provide business insights through dashboards.

4. Team Responsibilities

Member 1: Data Collection & Preprocessing - Gather pricing data - Handle missing values - Feature engineering
Member 2: Demand Prediction - Predict future demand - Analyze seasonal trends
Member 3: Dynamic Price Prediction - Regression models - Price optimization
Member 4: Competitor Analysis - Compare competitor prices - Market trend analysis
Member 5: Revenue Optimization - Calculate expected revenue - Recommend pricing strategies
Member 6: Dashboard & Frontend - Visualizations - User interface - Reports

5. Dataset

Suggested datasets: • Cab Rides Dataset • Weather Dataset • Competitor Pricing Dataset • Historical Demand Dataset
Features: Date, Time, Weather, Demand, Distance, Surge Multiplier, Competitor Price, Revenue, Holiday Indicator, Traffic Conditions

6. System Architecture

Data Collection → Data Preprocessing → Feature Engineering → Demand Prediction → Dynamic Price Prediction → Competitor Analysis → Revenue Optimization → LLM + RAG Layer → Streamlit Dashboard

7. Machine Learning Models

Traditional Models: • Linear Regression • Random Forest Regressor • XGBoost (Optional)
Advanced Model: • Neural Network using TensorFlow/Keras
Neural Network Architecture: Input Layer → Hidden Layer 1 → Hidden Layer 2 → Output Layer
Activation Functions: ReLU
Optimizer: Adam
Loss Function: Mean Squared Error

8. LLM Integration

The project must include an LLM. Options: • OpenAI API • Gemini API • Local Ollama Models (Llama 3, Mistral, Gemma) Capabilities: • Explain pricing decisions • Answer business questions • Generate pricing reports • Summarize market trends

9. RAG Component

Knowledge Base: • Pricing Policies • Business Documents • Market Reports • Historical Pricing Strategies Workflow: Documents → Chunking → Embeddings → Vector Database → Retrieval → LLM Response

10. User Questions Supported

1. What is the recommended price today? 2. Why did the price increase? 3. Why did demand decrease? 4. What are competitor prices? 5. What is expected revenue? 6. Show seasonal demand trends. 7. Which feature impacts pricing most? 8. What pricing strategy should I use? 9. What is tomorrow's demand forecast? 10. Generate a pricing report.

11. Dashboard Features

• Demand Forecast Charts • Dynamic Price Recommendations • Revenue Analysis • Competitor Comparison Dashboard • Feature Importance Visualization • Neural Network Performance Metrics • AI Assistant Chat Interface

12. Evaluation Metrics

Demand Prediction: • MAE • RMSE • R^2 Score Classification Components (if used): • Accuracy • Precision • Recall • F1 Score

13. Expected Outcomes

• Improved pricing decisions • Increased revenue • Better demand forecasting • Reduced manual effort • AI-powered business intelligence

14. Future Enhancements

• Reinforcement Learning for pricing • Real-time market data integration • Multi-agent pricing assistants • Cloud deployment • Mobile application support

15. Conclusion

The Dynamic Pricing System combines Machine Learning, Neural Networks, LLMs, RAG, and modern dashboards to create a complete AI-powered pricing platform capable of real-time decision making and business optimization.